



THE UNIVERSITY *of* EDINBURGH
Edinburgh Medical School

Tutorial 1

Data Types and Structures in Python and R

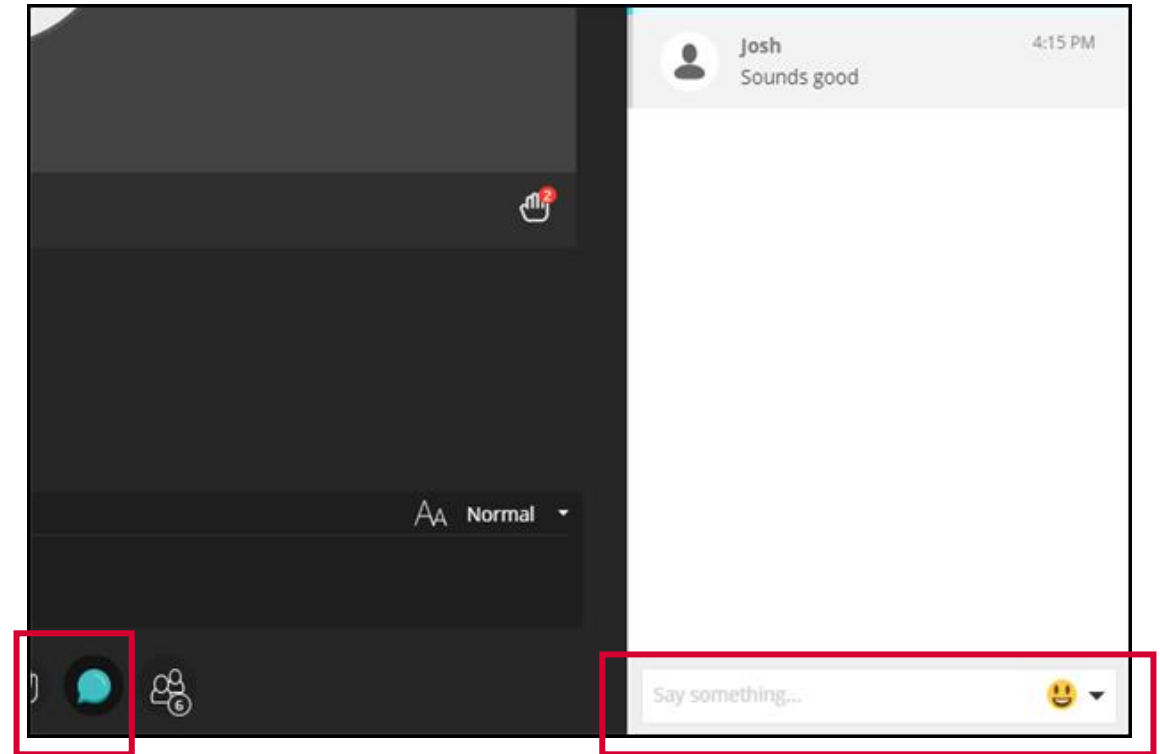
Audio check

Can you hear the presenter talking?

Please type **yes** or **no** in the “Text chat area”

If you can't hear:

- Check your Audio/Visual settings in the Collaborate Panel
- Try signing out and signing back into the session
- Type into the chat box and a moderator will try to assist you



Recording

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 Start Recording

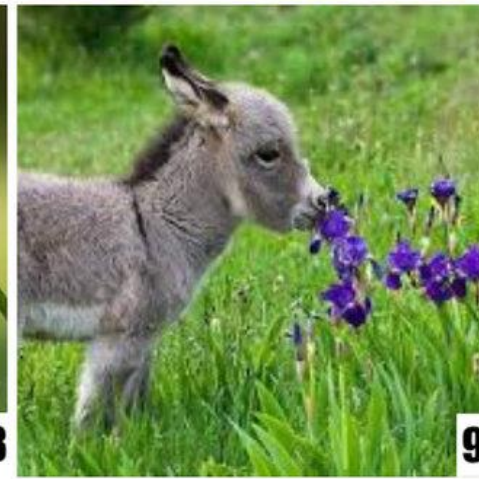


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Tutorial 1

Data Types and Structures in Python and R

How are you feeling today on this range of expressive animals?



Agenda

- Teaching team introduction
- Course overview
- Course GenAI Policy
- Accessing tutorial coding notes and materials
- Error of the week: quick note on a Noteable R warning message
- Practical: Python and R workbooks



The teaching team



And Richard!

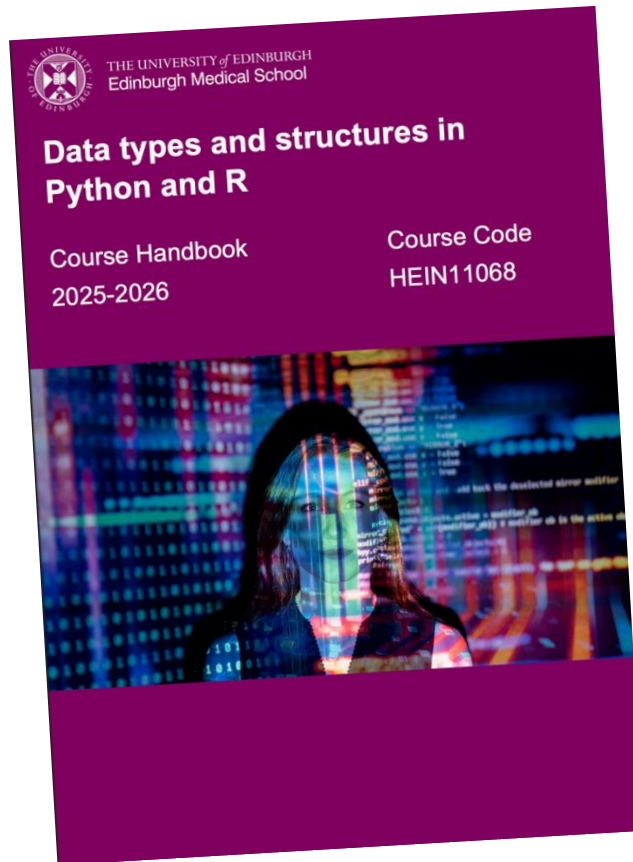
Course overview

- Live tutorials **Wednesday 4pm or Friday 10am BST**
- Drop-ins TBC (Week 3, Week 6)
- Content released each week on Friday
- 3 stars, a wish, and a step mini diaries
- Error of the week
- Assessment released in Week 4 – coding & written components
- Assessment due **29 June at 2pm BST**

Week	Week Commencing	Topic	Collaborate sessions
1	18 May 2026	Software Introduction	• Tutorial 20/05/2026 16:00 or 22/05/2026 10:00
2	25 May 2026	Data Types (part 1)	• Tutorial 27/05/2026 16:00 or 29/05/2026 10:00 • Drop-in TBC
3	1 June 2026	Data Types (part 2)	• Tutorial 03/06/2026 16:00 or 05/06/2026 10:00
4	8 June 2026	Essential Data Structures	• Tutorial 10/06/2026 16:00 or 12/06/2026 10:00
Assessment released			
5	15 June 2026	Advanced and Less Common Data Structures	• Tutorial 17/06/2026 16:00 or 19/06/2026 10:00
6	22 June 2026	Course completed; time allocated to work on final assignment	• Drop-in TBC
7	29 June 2026	Assignment: Submission deadline 29 June 2026	-----

Course AI policy

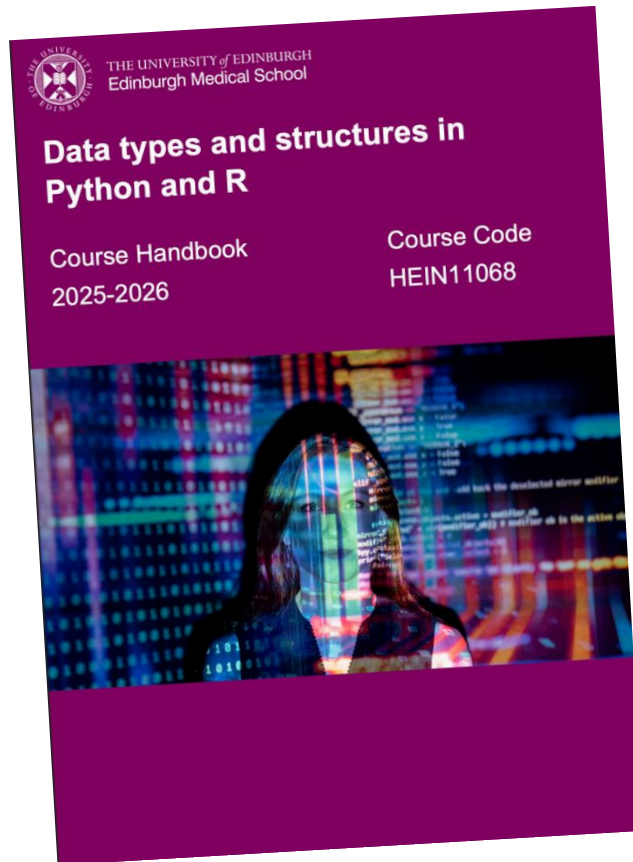
- Read the course handbook!
For the course AI policy and more!



- Accuracy, bias, and ethical concerns with any LLM use must be considered and personally weighed
 - Reminder: LLMs are probabilistic ***not deterministic/factual***
- Not banned but encourage you to use the human expertise available on the course (your teaching team!)
- At this level, reliance on LLMs **will have severe negative impacts** on any aspirations of technical competence

Course AI policy

- Read the course handbook!
For the course AI policy and more!



- Not permitted/acceptable to have generative AI do your work for you
 - Academic misconduct to represent AI generated content as your own work
- AI declaration **required** in your final assignment
- All submitted work (discussion boards, assignments) should be typed/written by you, your fingertips hitting the keys on your keyboard to produce your submission

✅ ACCEPTABLE - You ask generative AI to explain an error message to you.

❌ NOT ACCEPTABLE - You ask generative AI to write code for you.



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Please select a personal notebook server

Standard Python 3



Start

- Use to code in Python
- Use to access the files in your Noteable
 - Noteable runs on Jupyter in the background so the Jupyter Notebook File Browser is your home file directory

See Using Python in Noteable video in Week 1 Topic 2



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Please select a personal notebook server

RStudio



Start

- Use to code in R
- Can import and download files through the files pane, but more steps involved than in the Notebook File Browser

See Using R in Noteable video in Week 1 Topic 3

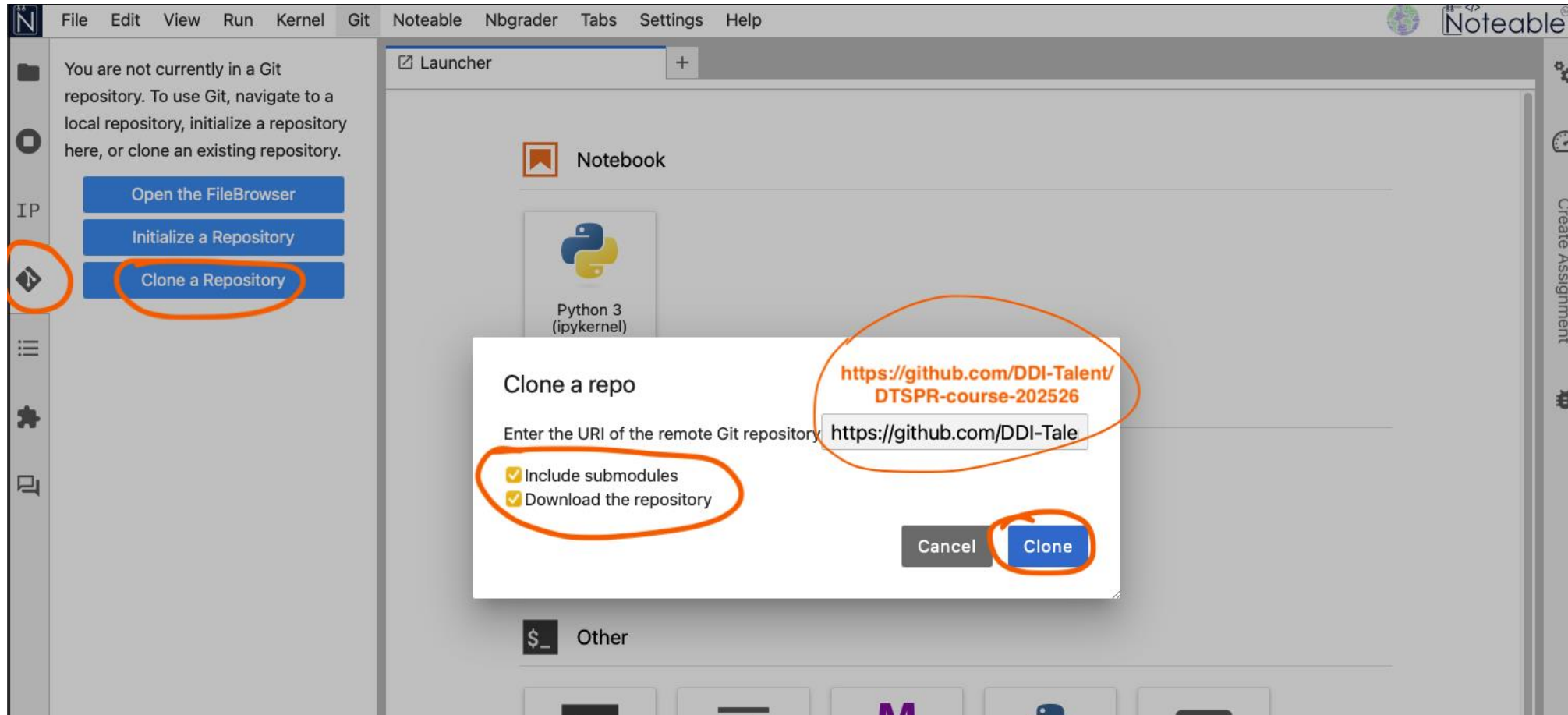
How to put files on your computer into your Noteable file directory

1. Download or identify files of interest
2. Open a Standard Python 3 server
3. [optionally] Launch Jupyter Notebook File Browser (from Help tab)
4. Upload the selected files
5. Have fun coding!

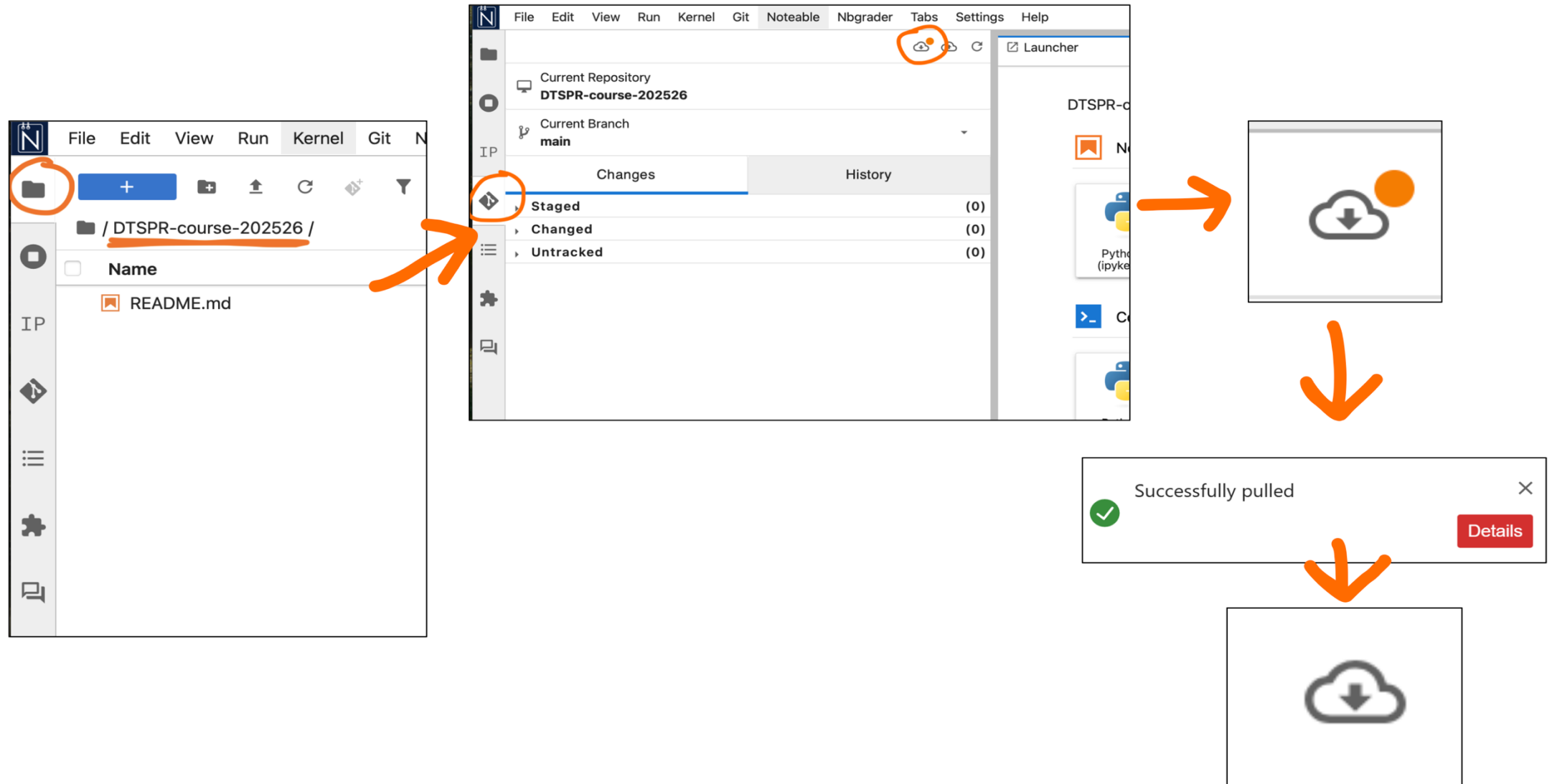
1st time accessing course notes & materials – “Clone” GitHub Repository



<https://github.com/DDI-Talent/DTSPR-course-202526>



All other times – “Pull” GitHub Repository



Our first error of the week

Which is a warning not an error, technically

Occasional Noteable RStudio warning message (& how to fix it)

- Nothing to be afraid of!
- Noteable is on the cloud, timezone is not guaranteed
 - Tidyverse can do all sort of cool date things (which we will see next week), so it needs to know the time

```
34 {r}
35 library(tidyverse)
36

Registered S3 methods overwritten by 'dbplyr':
  method      from
print.tbl_lazy print.tbl_sql
System has not been booted with systemd as init system (PID 1).
Can't operate.
Failed to create bus connection: Host is down
Warning in system("timedatectl", intern = TRUE) :
  running command 'timedatectl' had status 1
— Attaching packages — tidyverse 1.3.2 —
✓ ggplot2 3.3.6      ✓ purrr 0.3.5
✓ tibble 3.1.8       ✓ dplyr 1.0.10
✓ tidyr 1.2.1        ✓ stringr 1.4.1
✓ readr 2.1.3        ✓ forcats 0.5.2
— Conflicts — tidyverse_conflicts() —
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag() masks stats::lag()
```

Solution! Add this line of code first

```
22 {r libraries}
23 Sys.setenv(TZ="Europe/London") # set timezone (TZ argument) to whatever timezone
    you may be in, I am in the UK so am using the TZ identifier for London. The allows
    the to packages to work as intended
24
25 library(tidyverse)
26 |
27 ~~~
28
```

- You can use a TZ identifier as input from the area you are in
 - See [this Wikipedia page](#) with a list of time zone identifiers from the tz database

Now for some hands-on practice with
Jupyter Lab & Rmarkdown

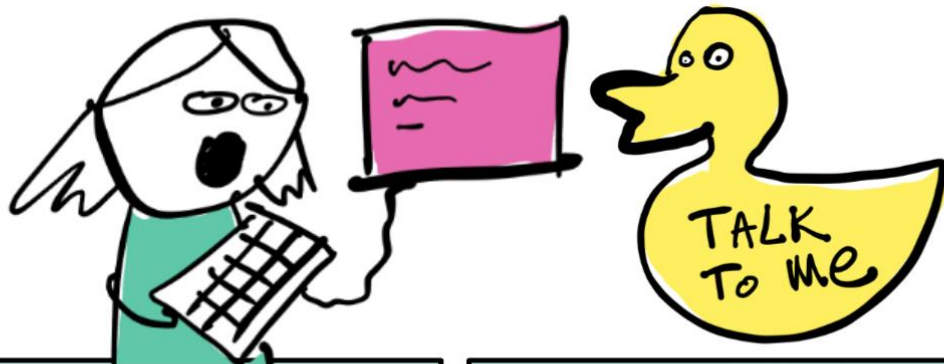
The tutorial documents have a lot of text to go
through, be sure to read carefully!

Pair Programming

programmers talking
through their
code together
and taking turns



Pair Programming



the Driver = active person,
problem-solving, thinking out loud

the Navigator = reflective person,
looks out for bugs, is a Rubber-Duck*



the Driver = active person,
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